
Specialisation and syndication as risk management strategies for venture capital firms in India

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Abstract: This paper presents an analysis of syndication and domain specialisation strategies pursued by the venture capital (VC) firms in India. Using the theoretical lens of the resource-based view, we explore how three main resource-related attributes: resource structure of the VC firms, resource requirements of the investee ventures and the ease of access to resources, drive the intensities of syndication and specialisation for the VC firms under study. We use the K-means cluster analysis technique, to analyse and profile four distinct VC firm segments: a) low syndication and low specialisation; b) high syndication and low specialisation; c) low syndication and high specialisation; d) high syndication and high specialisation. Our study contributes to the extant literature on VC investment strategies, top management teams and fills an important gap on the literature pertaining to VC firms in India.

Keywords: venture capital; syndication; specialisation; India; social capital; human capital; top management teams.

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1 Introduction

Venture capitalists (VCs) are financial intermediaries focused on funding projects in emerging high-technology realms. Nascent technologies, domains and business models and intangibility of assets (Gompers and Lerner, 2004), the mainstay of such VC funded projects; entail an extreme level of information asymmetry, which warrant specialised risk assessment skills. Adverse selection (risks from hidden information) and moral

hazard (risks from hidden actions) are the two prominent risks arising out of the asymmetric information between the VCs and the entrepreneurs (Amit et al., 1998). As such, the VCs are known to deploy several strategies to control such risks – with specialisation and deal syndication being the most prominent among these (Dimov and De Clercq, 2006).

In this paper, we focus on the specialisation and syndication strategies for the VCs in India. In general, VCs are unlikely to be generalists but rather, tend to specialise by the funding stage, industry, geography and even the deal size (Hochberg, et al., 2015a, 2015b; Buchner et al., 2017). Specialisation facilitates access to networks, information and deal flow, consequently providing a better handle on the magnitude of prevailing risks (Gompers et al., 2009). Syndication refers to the joint investment in a deal by two or more VCs with the aim of pooling their tangible and non-tangible resources (Wright and Lockett, 2003; Jääskeläinen, 2012). This resource-sharing not only aids in the selection of superior quality deals but also assists in monitoring the same efficiently (Hochberg et al., 2007; Dai and Nahata, 2016). To sum up, the value proposition of specialisation is access to ‘tacit knowledge’, a specialised resource in itself; while that of syndication is the enhancement of the pre-existing resource endowment by pooling the same across peer VCs – both of which are equally important for effective risk management (Wright and Lockett, 2003; Newbert, 2007).

Of late, India has established itself among the top destinations for the deployment of global VC funds (Bain & Company, 2014). The prominent foreign private equity firms such as Accel Partners, Sequoia Capital, Tiger Global, Softbank and so on have established their bases here, where they share space and compete for deals with the local VC firms. As such, the domestic and foreign VCs (DVCs, FVCs) are seen to exhibit different approaches to syndication and specialisation, which in turn may be possibly attributed to their differential areas of expertise and resource composition. In India, although the local VCs understand the native business landscape well, they exhibit limited opportunity recognition potential given their limited specialisation. In fact, one of the prominent examples of opportunity recognition failure is the rejection of funding a major IT services firm, Infosys Technologies by a local VC firm ICICI ventures. Infosys later grew to a scale and went on to successfully list on NASDAQ and NYSE (Dossani and Kenney, 2002). Other tech-focused DVCs that do well on opportunity recognition, encounter the challenge of limited financial resources. Thus, they view syndication as a conduit for financial resource augmentation. On the contrary, the FVCs encounter a slightly different challenge. These are endowed with abundant capital, but find the limited deal flow a major constraint while investing in India. Thus although, they wish to specialise, they end up being generalists (Joshi, 2015; Joshia, 2018a, 2018b forthcoming). Moreover, the ventures that address the core Indian problems of affordable healthcare, clean energy, agricultural innovations and so on are more likely to be located in second-tier cities. In such cases, access to local networks, understanding of local language and nuances of the local culture becomes a major impediment for FVCs. (The readers need to be reminded here that India by itself is home to at least 15 official languages and multiple sub-cultures). Under such circumstances, syndicating with local VCs, in order to piggyback on their networks becomes crucial.

To sum up, specialisation and syndication strategies are almost always used in conjunction. Since managing portfolio risks warrants the usage of both these strategies – assessing the joint role of syndication and specialisation strategies becomes important. While relative usage of the two strategies is unlikely to be uniform across the concerned

VCs, there could also be present possible interactions among the two based on the underlying resource composition. Thus, the fundamental resource structure that prompts the VCs to rely on one strategy more intensely as compared to the other becomes an interesting issue to examine in the context of VC investing. The primary research question we address in this paper is: *How does the underlying resource composition impact the variation in the relative intensities of syndication and specialisation (and the interactions therein) by the concerned VC firms?*

In particular, this study makes three broad contributions to the literature. First, we add to the literature on VC investment strategies – specialisation and syndication. While specialisation and syndication have been separately analysed (Hochberg et al., 2007; Dimov and De Clercq, 2006; Hochberg et al., 2015a, 2015b), the way their specific intensities and the interactions therein are driven by – the underlying resource structure of the VCs, the need to augment the same and the relative transactions costs of accessing such resources, has not been fully explored. Especially, issues such as ‘how the native institutional set-up hinders domain specialisation’ or ‘why is stage-specialisation easier to practice than sector-specialisation, especially in emerging economies’ is not yet well-understood. Similarly, the mechanisms other than syndication, which the FVCs rely on to augment their resources while investing in emerging economies are not understood well enough. This study furthers the existing knowledge pool in these areas.

Second, we contribute to the literature on the human capital of VC firms. While the impact of team composition on the alliance formation process and domain specialisation has been examined for other types of firms (Ireland et al., 2002), the same has not been sufficiently analysed in the context of VCs (Patzelt et al., 2009). Especially, how and why the specific skill sets of the team members such as erstwhile entrepreneurial or investing experience impact the VC firm’s inclination to use one strategy more intensely vis-à-vis the other needs to be better understood. This study aims to enhance the understanding of the same.

Finally, we extend the literature on VC investment strategies on emerging economies to India. While the investment strategies pursued by the VCs investing in China have been briefly examined (Huang et al., 2013; Gu and Lu, 2014), the same does not hold true for India. China and India are vastly different in terms of their institutions, political structure and policy making and hence the results from China cannot be simply extrapolated to India, rather India needs to be examined separately in its own right. This study aims at the same.

Rest of the paper is organised as follows. Section 2 provides an overview of the VC industry in India. Section 3 presents the survey of literature and Section 4 the conceptual framework. This is followed by the discussion about the data and the methods of analysis in Section 5. The results are presented in Section 6 and discussed in Section 7. Section 8 highlights the managerial implications and the policy conclusions emerging from the study.

2 The Indian context

Over the past decade, VC has emerged as one of the prominent conduits for funding businesses in the emerging domains (Planning Commission, India, 2012). As of 2014, there were about 309 VCs operating in India (Venture Intelligence, 2014) that have

collectively funded more than 5,000 businesses (Bain & Company, 2014), with a compounded annual growth rate of about 30% (on VC investments) between 2010 and 2014 (SEBI, 2014). Several MNCs such as Intel, Qualcomm, SAP and Cisco have established their corporate VC arms to leverage the technologies developed by the Indian start-ups (Planning Commission, India, 2012). Of late, the Indian VC – ecosystem has been booming, with various entities such as angel investors, academic incubators, industry accelerators, and technology business incubators contributing to the deal flow of potential VC deals (Joshi and Satyanarayana, 2014; Subrahmanya, 2015). Over the period, Bangalore, Mumbai and Delhi have emerged as important Asian cities in terms of the high concentration of VC funded companies (SeedTable, 2014) and Bangalore has been identified among the top nine start-up hubs of the world (Pullen, 2013).

And yet, despite their growing influence, the VCs investing in India encounter remarkable challenges. To start with, India ranks 100 out of 158 on the *Ease of Doing Business* index developed by the World Bank Group (2017) and 85 out of 175 on the *Public-Sector Corruption* index (Transparency International, 2014). A recent study on VC investing in India, revealed the prevalence of a severe trust deficit between VCs and entrepreneurs (Panda and Dash, 2015) especially in early stage ventures. As legal processes are yet to catch up with the rapidly evolving VC-entrepreneurial ecosystem here, it can be difficult to enforce contracts (Panda and Dash, 2015). Furthermore, closing or winding up a business in India is a complex and time consuming process (Planning Commission, India, 2012).

Another area of concern pertains to the low depth of the market for viable deals in India. The depth of the market can be established by the volume of VC investments in a region – the average annual VC investments between the years 2006 and 2013 were USD 31.2 billion in the United States as compared to that of USD 1.23 billion for India (Ernst & Young, 2014). The lack of depth of the Indian VC market potentially inhibits the establishment of highly specialised VC firms. Domain specialisation is also known to enhance the opportunity recognition potential (Shane, 2000) which in turn enables the VCs to tap into the potential investee firms in their very nascent phases itself. By hindering this process, low specialisation deters the process of discovery of such viable deals – especially in emerging technology domains. Likewise, it also results in the phenomenon of too much money chasing too few deals resulting in an over-valuation, translating to high entry level prices and consequently to lower rates of return – as can be seen in the case of retail e-commerce sector in India (VC Circle, 2014).

Collectively, these factors considerably enhance the investment risks for VCs in India which need to be tackled by the means of appropriate risk management strategies. It is in this context that the study of syndication and specialisation strategies becomes vital.

3 Survey of literature

The aim of this section is to provide an overview of the resource-based view (RBV) in the context of VC investing, exploring the strategies of specialisation and syndication and above-all understanding how the resource structure of VCs, need to augment the same and the relative ease of access to resources determine the VC firms' intensity strategy usage.

Per RBV, firms comprise a historically determined collection of resources tied semi-permanently to their management (Lockett and Thomson, 2001). Such resources,

fundamental to the firm's competitive advantage are valuable, rare, imperfectly imitable, and non-substitutable (Barney et al., 2001). Similarly, in case of the VC industry as well, the firm-level resource composition is extremely important. Moreover, as compared to other types of firms, VCs are much smaller in size (in terms of their human resources), and yet possess experience that can be considered both extensive and diverse (Dimov and Shepherd, 2005).

Effective VC investing is largely about appropriately exploiting the firm-level networks (Gu and Lu, 2014). In fact, VCs are rarely known to invest in ventures completely unknown to them (Joshi, 2015). The entrepreneurial-VC networks, the angel investor- VC firm networks, inter-VC firm and inter-entrepreneurial networks and those between incubators/business accelerators and the VC firms all play a vital role in guarding the VC firm from unpredictable risks arising out of information asymmetry (Joshi, 2005; Mainela and Puhakka, 2011). The resources of the VC firm are essentially the conduits for establishing such networks (Joshi and Pruthi, 2017). Naturally, the manner in which they devise strategies to facilitate their access to resources from the viewpoint of risk management becomes an important arena of study. In fact, syndication and specialisation maybe viewed as strategies that enable the VC firm to augment their pre-existing resource pool (Joshi, 2015).

3.1 Specialisation and syndication

Tacit knowledge is a significant source of competitive advantage for firms, especially in domains with relatively large information asymmetries (Meihami and Meihami, 2014; Ahmad et al., 2014). Per the *resource-based theory*, domain-specialisation is understood to equip the VCs with *idiosyncratic* or tacit knowledge which enables them to appropriately assess the underlying risks (Barney et al., 2001; Newbert, 2007). Such, domain specialists are known to be the most responsive to signals of viable investment opportunities (Gompers et al., 2009) and are hence likely to spot better quality deals, thus reducing the magnitude of adverse selection risks (Gompers et al., 2009). In the VC industry, where the information on viable deals is seldom public, some of the best deals are necessarily the ones that are proactively pursued by the VCs themselves (Hsu, 2007; Joshi, 2015). However, getting a selective access to such deals much before the same are revealed to one's competitors warrants the presence of significant social capital in the form of networks of VC professionals (Sorenson and Stuart, 2008). Access to such networks and accordingly the deals is possible only when the firm is highly specialised in the concerned domain (Hochberg et al., 2015a, 2015b). For innovation-driven businesses with a strong prevalence of the first-mover advantage (Thornhill, 2006), such networks also serve as significant barriers to entry for newer VCs (Frynas et al., 2006; Hochberg et al., 2010).

The tacit knowledge arising from domain specialisation likewise enables them to keep a check on the risks emanating from moral hazard. By providing a better insight into the specific complexities associated with certain investment stages or industries, it enhances the scope for cross-sectional and serial knowledge spill-overs (De Clercq and Arenius, 2006). The specialised VCs also benefit from the learning curve effects which results in an accumulation of superior knowledge over time (Hochberg et al., 2015a, 2015b). Collectively, these factors enhance the level of control by VCs, thus guarding against the opportunistic behaviour of the entrepreneurs (Hochberg et al., 2015a, 2015b).

Syndication is a form of inter-organisational co-operation that serves the purposes of financial intermediation as well as achieves the specific objectives of individual firms (Jääskeläinen, 2012). As such there exist multiple motives of syndication. The finance-based motivation propounds syndication to arise from the need for risk diversification. For VCs with smaller fund sizes, syndication is probably one of the most potent ways to reduce systematic risks by allowing for investment in larger number of diversified companies (Robbie and Mike, 1998; Cumming, 2006). Deal flow motivation is another important motive of syndication. By granting access to a greater quantity and superior quality of deals (Sorenson and Stuart, 2001), it facilitates in the selection of better quality deals thus significantly reducing risks from adverse selection.

The value-add motive views syndication as a means of adding value – during both *ex-ante* and *ex-post* investment phases. During the *ex-ante* phase, pooling resources across VCs enables them to arrive at a clear-cut judgment of deal quality (Dimov and Milanov, 2010; Casamatta and Haritchabalet, 2007). The pooling of expertise and support during the *ex-post* investment phases, helps in alleviating the magnitude of agency risks and the consequent value-add helps in enhancing the overall valuation of the investee venture (Sorenson and Stuart, 2001; Bruining and Wright, 2002; Meuleman et al., 2009).

3.2 Resource based view in the context of VC investing

There are three principal aspects with regards to resources that are vital here – one the pre-existing resource endowment of the VC firm, two – the need to access greater resources and three – the ease of access to resources.

3.2.1 Pre-existing resource endowment of the VC firm

The human capital of the VCs refers to the size and composition of its core team. In fact, VCs earn their niche because of the tacit knowledge they possess to evaluate and manage projects with high information asymmetry (Amit et al., 1998). Such tacit or idiosyncratic knowledge does not exist independently but is rather endowed in the skill sets of the VC investment executives (Dimov and Shepherd, 2005). In general, larger teams imply more resources, a greater range of analytical skills and accordingly a greater spread of networks (Lockett and Wright, 2001). Thus, the presence of large investment teams decreases the need for pooling additional complementary resources and accordingly reduces the reliance on syndication.

The specific skill sets of the VC professionals are important as well. Prior founding experience of VC professionals has been found to enrich their perceptions of the underlying risks, enhance their knowledge of market entry and augment their potential for identifying agency problems (Patzelt et al., 2009). They are also more likely to understand the underlying nuances of the businesses especially at earlier stages or in certain domains and thus are more likely to practice domain specialisation as a risk management strategy (Patzelt et al., 2009). Similarly, those with prior experience in the VC industry are likely to be well networked within the VC community and possess greater social capital. These traits in turn influence the nature of strategies deployed by them. It has been seen that 'generalist' VCs (i.e., VCs with basic know-how in financing new enterprises) often trade their 'access to deal flow' for 'access to knowledge' by co-investing with 'specialist' VCs (those with specialised industry or founding

experience) which in turn gets reflected in their higher intensity of syndication (Jungwirth and Moog, 2004).

The other vital resource of the VC firm is its financial capital. In general, higher financial resources reduce the need to syndicate for risk diversification purposes and thus VCs with relatively higher funding sizes are likely to syndicate less intensely (Jungwirth and Moog, 2004). The impact on specialisation would largely depend on sectors that these VCs are likely to concentrate on. Certain sectors such as drug discovery or E-commerce are known to be highly capital-intensive ones and thus will naturally result in a narrowing down of focus and thus specialisation (Joshi, 2015). On the contrary, if the industries are less capital intensive, there is a scope to support a wider range of ventures with the same level of financial resources. Moreover, for VCs with large fund sizes, the ability to specialise also depends on the viable deal flow at the investment destinations. In general, markets that lack depth are likely to significantly deter specialisation (Cumming, 2006).

The social capital of the VCs is another vital resource that influences its intensity of specialisation and syndication. The firm-level experience represented by the age of the VC firm and by the quantum of historically funded deals is an important proxy for status, reputation and accordingly social capital (Lu et al., 2013). In general VCs with a lower level of experience possess less social capital (Sorenson and Stuart, 2001) and are found to syndicate more in order to piggy-back on their more experienced counterparts (Hopp and Rieder, 2011) for the same. By the same rationale, the VCs with a greater level of social capital are also more likely to be approached by other VCs for potential syndication relationships and are thus likely to syndicate to a much higher extent (Dimov and De Clercq, 2006).

The impact of social capital on specialisation is however ambiguous. It has been often seen that VCs seek co-investment partners similar to themselves in terms of their reputation and status, investment focus and experience since the same helps them alleviate the uncertainty related to new partners (Trapido, 2007; Du, 2009). Doing so is likely to enhance the level of domain specialisation for the focal VC firm. However, other studies reveal that VCs choose syndication partners with distinctly different arenas of specialisation, while diversifying into newer sectors (Joshi, 2015). Thus, the evidence of the direction in which social capital impacts specialisation is far from conclusive.

The sector focus of the VC firm is another important attribute that determines its strategy usage. Overall, as compared to the 'generalist' VCs, sector-focused VCs are likely to better understand the nuances of the businesses that they invest in (Jungwirth and Moog, 2004). Moreover, the quality of their learning outcomes is further enhanced when the firm assimilates new knowledge that is related to its existing knowledge structure. These attributes enhance the possibility of sector-focused VCs to be specialists in the same domains.

3.2.2 Need to access greater resources

The nature of ownership of the VC is an important proxy for its resource structure. FVCs are known to possess low social capital at the investment destinations (Schertler and Tykova, 2011; Dai et al., 2012). By this rationale, the FVCs are more likely to syndicate in order to overcome their 'liability of foreignness'. In fact, in India, since its very inception, the FVCs have consciously pursued the strategy of hiring their senior executives from the DVCs with the aim of piggy-backing on the latter's social networks

(Dossani and Kenney, 2002). The pre-existing social capital of these individuals facilitates the syndication process and is likely to positively impact the overall syndication intensity. They are also less likely to be specialised especially with respect to earlier investment stages or high-tech domains as such ventures are often riskier, owing to another layer of information asymmetry (Dai et al., 2012).

High-tech sector-focus of the VC firms also impacts their need to access greater resources. The level of information asymmetry and accordingly the risks arising from the same are quite high for high-tech sector focused VCs, owing to the informational opaqueness and intangibility of assets (Dai et al., 2012). Thus, the need for second opinion is greater and consequently, the high-tech focused VCs are likely to be associated with a high intensity of syndication (Sorenson and Stuart, 2001). Also, evaluating ventures in the high-tech domains in the *ex-ante* phase necessitates the presence of specialised expertise, which can be obtained only from prior experience in the same domain (Patzelt et al., 2009). Thus, the VCs evaluating such ventures are also likely to be highly specialised.

3.2.3 *Ease of access to resources*

The physical location of the VC firm is also important since it is likely to influence its access to resources. Location in a prime entrepreneurial hub with a vibrant VC-entrepreneurial ecosystem, facilitates access to networks, human capital and other tangible resources (Sorenson and Stuart, 2008; Patzelt et al., 2009; Zarutskie, 2010). Frequent entrepreneurial meet-up events, hackathons and free movement of human capital across firms, entrepreneurial spawning and moreover the presence of a strong start-up culture (Mason, 2007) can significantly enhance the prospects of syndication opportunities. Abundant availability of specialised human capital is also likely to enhance the intensity of specialisation (Qian et al., 2012). Specialised VC firms are also more likely to establish themselves close to university set-ups in order to ease access to the deal flow from academic incubators. The congregation of many VCs in the same location further aids syndication (Joshi, 2015).

To overcome agency risks, VCs need to establish offices as close as possible to the physical location of the ventures funded by them. Surprisingly, this is common even in today's era of where information can be seamlessly shared across digital corridors (Lutz, et al., 2013). That's the reason we find examples of such geographical concentration of VCs and start-ups such as – Silicon Valley and Boston route 120 in the US, Cambridge in UK and Bangalore in India. Given the fact that the deal flow might not be localised but is rather spread across multiple geographic clusters (especially in non-high-tech domains), one way of addressing the issue of physical proximity is to set up VC offices at multiple locations. This enables them to piggyback on the niche regional networks in each distinct geography (Joshi, 2015). Of course, it's not possible for all VCs to do so, but rather for only those with a sufficient scale. However, this strategy could possibly reduce the reliance on syndication with the VCs in other locations.

3.3 *Research gaps*

While syndication and specialisation strategies have been extensively examined earlier (Hochberg et al., 2007; Dimov and De Clercq, 2006; Hochberg et al., 2015a, 2015b), the underlying factors that result in the interactions among the two are yet to be suitably

explored. In general, VCs that demonstrate high syndication intensity need not be uniform in terms of their intensities of specialisation. Similarly, among the domain specialised VCs as well, there could possibly exist firms with varying intensities of syndication. Thus, although the VCs seem overtly homogenous in terms of the usage of one strategy they could distinctly differ in terms of the other strategy used by them.

Especially, factors such as ‘how and why the native institutional set-up hinders domain specialisation’ or ‘why is stage-specialisation easier to practice than sector-specialisation, especially in emerging economies’ is not yet well-understood. Similarly, the mechanisms other than syndication, which the FVCs rely on to augment their resources while investing in emerging economies are not understood well enough. Moreover, how the mode of foreign entry of FVCs impacts their strategy usage is also not well-assessed.

There exists only limited literature so far in the context of human capital of VC firms. While the role of human capital in impacting the process of alliance formation has been probed for other types of firms viz. bio-technology, high-tech, hospital set-ups, energy, and banking (Krishnan et al., 2006; Lee and Park, 2008), their specific role in impacting the alliance formation and also domain specialisation in the VC industry in the form of syndication needs to be further analysed. Currently, there are very few studies in this domain (Dimov and Shepherd, 2005; Patzelt et al., 2009; Joshi and Pruthi, 2017). Especially, how and why the specific skill sets of the team members such as erstwhile entrepreneurial experience or prior VC investing experience impact the VC firm’s inclination to use one strategy more intensely vis-à-vis the other has not been examined.

Prior studies have explored syndication strategies of VCs in China (Wright, 2007; Gu and Lu, 2014; Lu et al., 2013). Although outwardly similar in many respects, India and China are strikingly different in terms of political systems, legal and financial institutional frameworks, innovation models, and overall attitudes towards free enterprise (Huang and Khanna, 2003). Thus, the results from China cannot be simply extrapolated to India rather India needs to be examined separately in its own right.

4 Conceptual framework

Based on the survey of literature and the research gaps identified therein, we propose the following conceptual framework.

Per this framework, the range of investment strategies pursued by the VC firms is driven by three sets of factors – resource endowment of the VC firm (viz. the financial, human and social capital); the resource requirements of investee firms and the ease of access to resources. Together, these factors determine the variations in the intensities of syndication and specialisation.

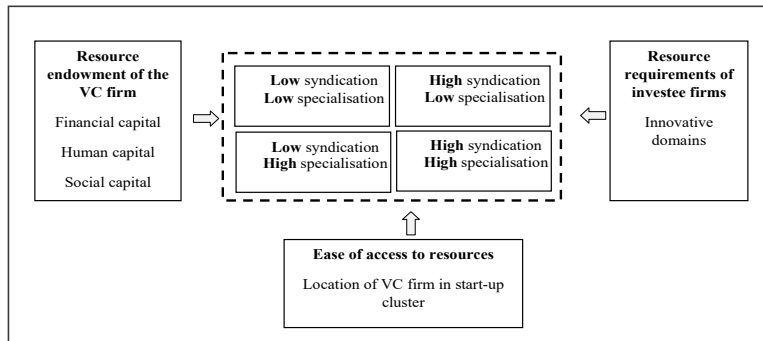
From the framework, we derive the following propositions:

- Proposition 1 Higher level of financial capital lowers the intensity of syndication and specialisation.
- Proposition 2 Specific composition of human capital and not merely its size is important in determining the syndication and specialisation intensities.
- Proposition 3 Higher level of social capital is vital in impacting syndication but not necessarily specialisation intensity.

Proposition 4 Ease of access to resources enhances both syndication and specialisation intensities.

Proposition 5 Innovative businesses warrant a high intensity of both syndication and specialisation.

Figure 1 Investment strategies and resource structure



5 Methodology

5.1 Scope, data and methods of analysis

This study is mainly based on secondary data obtained from venture intelligence (VI) – a private database on the VC and PE deals in India. In addition, we interviewed about eight VCs with the aim of validating the statistical results obtained from this exercise. Additionally, we have also used information obtained from the respective VC firm websites after seeking relevant clarifications with the concerned VC professionals.

The unit of analysis for the study is an individual VC firm. We have used a sample of 70 VC firms in this analysis. We regard our sample to be representative since the VCs in our sample have funded about 80% of the deals in India between 2007 and 2013 (Venture Intelligence, 2014). Although, there were about 309 VCs firms in the VC firm directory, not all of them were active. Hence, in order to take care of the 'active status', we chose only those VCs that had funded at least three deals between 2010 and 2013. Thus, we narrowed down from 309 to 70 VC firms.

These VCs are spread across all major Indian cities – (Mumbai, Bangalore, Chennai, Delhi (NCR area), Kolkata, Hyderabad and Ahmedabad), while some of them also operate remotely from other countries as well (USA, UK and Singapore). For the analysis, we have considered the deals funded between the period 2005 and 2013 and suitably aggregated the same at the VC firm level.

Data analysis involved two steps: First, clustering the VCs in our sample based on their respective intensities of specialisation and syndication using the K-Means clustering algorithm. Second, profiling the VCs based on VC firm-level, portfolio-level and other ecosystem-related attributes. All analysis was performed using SPSS 21.0.0.0 software.

It must be clarified that ‘specialisation intensity’ has been defined based on the investment-stage (early-stage focus) of the VC firm. For defining the same, we could have used the sector-focus, geographic focus or funding size focus as well. However, we consciously decided against the same for the following reasons: prima-facie, it appeared that the Indian VCs were not very sector-focused. Based on the VI data, we found that on an average, each VC firm had invested in about six sectors. Therefore, we ruled out the possibility of sector-specialisation to start with. On funding-size, we did not have the data available for more than 25% of the deals in the sample. Even where it was available, we could get access to only the overall deal size and not necessarily the contribution made by the individual VC firm in each deal. Regarding the geographical location of the funded deals, we found there existed a skewed distribution (with more than 30% of the deals belonging to just one Indian city viz. Bangalore). Therefore, we chose to use just early-stage focus in the clustering scheme whereas others were used as profiling variables.

5.2 Variable definitions

5.2.1 Variables used in cluster analysis

- *Intensity of syndication*: this variable was defined by computing the ratio of the number of syndicated deals to the total number of deals for each VC firm.
- *Intensity of stage specialisation (by early investment stage)*: this variable was defined by computing the ratio of the number of early-stage deals to the total number of deals for each VC firm.

Since, in India, there does not exist an objective definition of ‘early-stage’ per se, for this analysis, we decided to use the definition from the VI database. (In the VI database, early-stage has been defined as first/second round of institutional investments into companies that are:

- 1 less than five years old
- 2 not part of a larger business group
- 3 investment amount is less than USD 20 M.

In our database, the average size of an early-stage funded deal is around USD 3.46 million.

5.2.2 Variables used in profiling clusters

5.2.2.1 Financial capital

The data pertaining to fund size of each of the VC firms were obtained from VI and other private websites that track the ongoing activities of the VCs operating in India (VCCircle.com, Crunchbase.com, Yourstory.com). The financial resources of the VC are represented by their respective fund-sizes in USD million.

5.2.2.2 Human capital related variables

The demographics of the VC team members were obtained from the respective VC websites. On the lines of Dimov and Shepherd (2005) and Patzelt et al. (2009), we visited the sections entailed ‘management’ or ‘investment team’ on websites, where the CVs of the team members are described. Following Patzelt et al. (2009), the VC professionals with the highest-ranking titles (such as – general partner, partner, principal, managing director, president, vice-president) were deemed to comprise their core investment team. We created three different variables:

- *Team size*: number of investment executives for each VC firm.
- *Prior founding experience (in India and overseas)*: indicator variable to determine if any of the investment executives have established a business venture during the past.
- *Prior experience in VC investing (in India and overseas)*: proportion of investment executives with prior experience of having worked for other VC firms in India or overseas

5.2.2.3 Social capital related variables

As the proxy for social capital, we create the following variables:

- *VC age*: this refers to the number of years of operations of the VC in India (difference in years – between the years 2014 and the year the VC was established in India).
- *Historically funded deals*: this refers to the aggregate number of deals funded by each VC since 2005.
- *Presence of offices at multiple locations in India*: an indicator that takes value 1 if the VC has offices across multiple locations in India.

5.2.2.4 Nature of ownership of the VC firm

We create two variables here.

- *Foreign VC*: indicator variable takes the value 1, if the concerned VC is of foreign origin.
- *Mode of foreign market entry*: this is an indicator variable that takes the value 1 in case the FVC has a physical office in India, 0 otherwise (The foreign market entry strategies of FVCs are likely to significantly differ. Some of them do not even establish a physical office in India and carry on their operations remotely from their respective countries of origin).

5.2.2.5 Sector-focus related variables

The variables in this category relate to the sectors of the investee firms funded by them. In this category, we create the following variables

- *Numbers of sectors of investment*: this variable indicates the number of sectors that each VC firm invests in. The sectors corresponding to each deal have been obtained from VI.
- *Herfindahl index*: based on Gompers et al. (2009) we compute the index of sector concentration. The value of this index is not merely based on the count of sectors but on the proportion of deals belonging to each sector – which gives a true measure of sector-concentration, e.g., If a firm invests in total 6 sectors but about 90% of its deals come from just two sectors out of these, the diversification measure based on the Herfindahl index would categorise the VC firm as ‘sector-concentrated’ while that based just on the *number of sectors of investment* will classify it as ‘sector-diversified’. The values of the Herfindahl index range between 0 and 1, with greater values indicating a higher level of sector concentration.
- *Proportion of high-tech deals*: for defining this variable, we compute the ratio of the number of deals belonging to high-tech domains to the aggregate deals for each VC. The deals belonging to IT and ITeS and Bio-technology sectors as defined in the VI database were classified as high-tech.
- *Sector-focus of the VC*: if greater than 50% of the deals of the VC are concentrated in any *single* sector (viz. IT and ITeS, Healthcare, Energy, Education, BFSI and so on), then the VC is categorised as sector focused and accordingly we create an indicator variable that takes the value 1.

5.2.2.6 VC-entrepreneurial ecosystem

To capture the impact of the ecosystem related factors, we create an indicator variable that takes the value 1 if the VC firm is located in Bangalore.

5.3 Sample description

About 88% of the VCs belong to just three main start-up clusters – Bangalore, Mumbai and National Capital Region (New Delhi, Gurgaon, Ghaziabad, NOIDA, Faridabad). The average age of these VCs is around six years and they have funded about 26 deals on an average. The average fund size is around USD 321 Mn. About 40% of the VCs are of foreign origin. 46% of the deals funded by the VCs have an early-stage focus and about 54% of all deals are syndicated. The VCs invest in about six sectors on an average.

6 Analysis of clusters

The purpose of this section is to present the statistical results obtained from the K-means cluster analysis procedure. K-Means is most commonly used non-hierarchical clustering algorithm and considered appropriate for social science research (Hennig and Liao, 2013). The numbers of clusters in this technique however need to be specified upfront. The initial number of clusters is often determined based on – either prior research or data from other multivariate analysis or based on the information about the basic character of these clusters (Hair et al., 2015). Since our primary objective was to assess the determinants of the varying intensities of syndication and specialisation and the

interactions therein, we chose a 4-cluster solution – wherein the clusters were defined as: low syndication and low specialisation, high syndication and low specialisation, low syndication and high specialisation and high syndication and high specialisation.

In this section, we present output obtained from cluster analysis – cluster membership, one-way ANOVA output for the input variables of the clustering procedure, the final cluster centres and Distances between final cluster centres and the cluster profile results based on one-way ANOVA and Chi-Square tests of independence.

Table 1 Cluster membership

<i>Cluster type</i>	<i>Cluster1 low syndication and low specialisation</i>	<i>Cluster2 high syndication and low specialisation</i>	<i>Cluster3 low syndication and high specialisation</i>	<i>Cluster4 high syndication and high specialisation</i>
Proportion of sample observations	28%	24%	24%	24%

From Table 1, the sample observations are fairly uniformly distributed across the four clusters.

Table 2 reveals that subjecting the concerned variables (viz. the proportion of syndicated deals and proportion of early-stage deals), to the K-means clustering algorithm, brings out four clusters with varying intensities of syndication and specialisation.

Table 2 One-way ANOVA results – cluster variables

<i>Clusters</i>	<i>Cluster1 low syndication low specialisation</i>	<i>Cluster2 high syndication low specialisation</i>	<i>Cluster3 low syndication high specialisation</i>	<i>Cluster4 high syndication high specialisation</i>
Proportion of syndicated deals	32%	79%	14%	63%
Proportion of early-stage deals	16%	27%	85%	87%
	<i>Overall sample</i>	<i>F-statistic</i>	<i>P-Value</i>	
Proportion of syndicated deals	54%	56.69	0.000	
Proportion of early-stage deals	46%	87.15	0.000	

Table 3 shows the Euclidean distances between the final cluster centres. Greater distances between clusters imply greater dissimilarities. From this, we can conclude that clusters 2 and three are most different and clusters 3 and 4 least different from each other in terms of their characteristics.

Having identified the clusters, the next task is to profile each of them based on their vital attributes. In Table 4, we present the profiling results that will enable us to further understand the underlying characteristics of each of the clusters. The clusters have been

profiled with respect to the following attributes – financial, human and social capital related, portfolio focus related and start-up ecosystem related.

Table 4 also gives the results from the one-way ANOVA or Chi-square tests of independence with respect to each of the profiling variables across the four clusters. Based on the statistical tests, the four clusters are found to be significantly different across most of the attributes. Even where they are insignificant, the results are directionally indicative.

Table 3 Distance between final cluster centres

<i>Clusters</i>	<i>Cluster1 low syndication low specialisation</i>	<i>Cluster2 high syndication low specialisation</i>	<i>Cluster3 low syndication high specialisation</i>	<i>Cluster4 high syndication high specialisation</i>
Cluster1 low syndication low specialisation		.483	.725	.786
Cluster2 high syndication low specialisation	.483		.862	.610
Cluster3 low syndication high specialisation	.725	.862		.491
Cluster4 high syndication high specialisation	.786	.610	.491	

7 Discussion of results

In this section, we first analyse the profiles of each of the clusters and then focus on explaining the underlying strategy interactions

7.1 Cluster profiles

7.1.1 Low syndication – low specialisation cluster

The VCs belonging to this cluster are known exhibit the lowest intensity of syndication and early-stage specialisation. An understanding of the cluster profile will enable us to appreciate the nuances of the strategies pursued by them. The VCs belonging to this cluster have largest fund size (around USD 440 MN) and thus seem to be relatively well-endowed with financial capital. They also rank the highest in terms of the size of their investment teams. Thus, they seem to be endowed with adequate human capital as well. The level of social capital can be assessed using various proxies. First, these firms are the oldest in terms of the number of years of operations in India (about nine years as compared to sample average of six years). Having spent considerable period at the investment destinations enables them to build a network of contacts and makes them better adapted to the local institutional environment (Meuleman and Wright, 2011).

Moreover, these VCs largely focus their investments on the more conventional sectors. In fact, this segment has the presence of lowest proportion of VCs with a high-tech domain focus (about 13% as compared to the sample average of 49%). Neither, are these VCs sector-focused in general. Rather, they are highly diversified, i.e., they invest in the highest range of sectors (9 sectors as compared to the overall average of 6). Also, the Herfindahl index value is very low indicating low industry concentration (0.276 as compared to the sample average of 0.540).

The strategies used by the firms in this cluster are clearly reflective of their underlying characteristics. In general, due to the low focus on high-tech domains and early-stage ventures, the magnitudes of information asymmetry risks are likely to be quite limited (Sorenson and Stuart, 2001; Dai et al., 2012). This reduces the need for pooling information and opinion across VCs at the investment decision-making stage. Adequacy of financial resources does not necessitate financial resource pooling for risk diversification purposes. In terms of their human resource composition as well, this cluster has the highest proportion of investment executives with the experience of having worked for other VC firms in India and overseas. Thus, the adequacy of their human resources does not warrant resource pooling for monitoring of value-add purposes as well. Since, these VCs are also the oldest in terms of the number of years of operations, the level of pre-existing social capital is also likely to be high, which enhances their decision-making proficiency as compared to their less experienced counterparts. In fact, previous literature has acknowledged the presence of significant decision-making differences among the experienced and inexperienced VC firms (Sorenson and Stuart, 2001; Butler and Goktan, 2007; Cumming and Dai, 2010) wherein the important proxy for experience is the VC firm's age (Sorenson and Stuart, 2001). Well-endowed resource structure, together with the confidence in adjudging deals arising out of their investing experience in investment destinations reduces the need for syndication.

The VCs in this segment are also less likely to be early-stage specialists. Early-stage specialisation makes sense when the focus of investments is high-tech domains. Acquisition of technology firms at earlier stages (when the valuation is much lower) often translates to much higher valuations at later stages (Patzelt et al., 2009). But the same rationale need not apply to the more conventional sectors. In fact, in certain sectors – there exists a late-mover advantage, wherein the late-entrant does not necessarily need to bear the risks that the first-mover has already encountered (Besharat et al., 2016). The later entrant can learn from the experience of the pioneer – thus enjoying lower costs and making fewer mistakes (Shankar and Carpenter, 2013). Moreover, quite often, the pioneers do not have deep pockets; thus, although they create a competitive business, they are unable to dominate it. So, competitors who enter quickly with more resources can win the business that the pioneer has created (Besharat et al., 2016). The VCs in this segment well-exhibit the above pattern. An executive from an important VC firm in this segment, which has funded one of the prominent online grocery portals in India, mentioned the following “our forte is to provide capital at the scaling up phases of the businesses. We do not wish to bear the pioneer risks – rather, we select the businesses with traction and provide adequate capital such that this in itself acts as a potential entry barrier for others.”

Table 4 Cluster profiles

	Low syndication low specialisation	High syndication low specialisation	Low syndication high specialisation	High syndication high specialisation	Overall sample	F-value/Pearson Chi-squared value	P-value
VC firm experience related variables							
Historically funded deals	26	32	11	34	26	1.748	0.166
VC firm age (in years)	9.11	7.38	3.63	4.31	6.24	5.888	0.001
Financial resources							
Fund size	440	164	20	30	120	5.888	0.001
Human resources related variables							
Team size	6.94	6.71	3.93	6.62	6.08	1.541	0.213
Proportion of VC firms (prior founding experience—in India and overseas)	5%	13%	38%	56%	27%	5.62	0.002
Proportion of VC firm professionals							
Prior experience in VC investing (in India and overseas)	94.7%	84.6%	66.7%	81.3%	83%	1.564	0.208
Portfolio focus related variables							
Proportion of high-tech deals	14%	54%	56%	75%	49%	7.708	0.000
Proportion of sector specialised VC firms	26%	75%	88%	94%	69%	11.307	0.000
Number of sectors of investment	9	5	3	5	6	6.989	0.000
Herfindahl index (for sectors of investment)	0.276	0.591	0.694	0.651	0.540	12.302	0.000
Other VC firm related variables							
Proportion of FVCs	42%	63%	19%	38%	40%	2.236	0.093
Presence of a physical office in India (FVCs)	88%	100%	33%	83%	76%	8.491	0.032
Proportion of VC firms with offices in multiple locations in India	16%	19%	19%	25%	19%	0.478	0.934
Ecosystem related variables							
Physical location—Bangalore	11%	53%	25%	50%	33%	9.647	0.022

7.1.2 High syndication – low specialisation cluster

The VCs in this segment have low intensity of early-stage specialisation but they demonstrate a high intensity of syndication. These VCs have a fund size of USD 160 MN (higher than the sample average). They have also funded the highest number of deals historically (about 32 deals as compared to the sample average of 26). The proportion of high-tech deals funded by them is also higher than the sample average (54% as compared to 49%). The cluster also represents an above average proportion of sector-focused VCs (75% as compared to 69%). Most importantly, this segment has the highest concentration of FVCs (about 63% of the VCs are of foreign origin as compared to that of 40%) in the overall sample. Also, 100% of these FVCs have a physical office present in India.

There could be many possible reasons for the high syndication intensity of the VCs in this cluster. One, their focus on later-stage ventures coupled with their relatively lower fund sizes. The deal size for later stage ventures is much higher and the only way to attain the same in the absence of one's limited resources is by means of syndication (Wright and Lockett, 2003). Two, an overwhelming proportion of the VCs in this segment are in Bangalore (53% as compared to the sample average of 33%). Bangalore being the start-up capital of India has a vibrant ecosystem in terms of the presence of VCs, incubators, accelerators, business angels and a critical minimum mass (Joshi and Satyanarayana, 2014; Subrahmanya, 2015). Thus, co-location with other elements of the ecosystem possibly facilitates the access to resources that these VCs lack thus translating to higher syndication.

The third reason for high syndication could be attributed to the overwhelming presence of FVCs in this cluster. In general, FVCs are known to possess lower social capital in the investment destinations (Guler and Guillen, 2010; Delloite, 2010; Dai et al., 2012) which they can overcome by forging co-investment relationships with DVCs or other FVCs with a greater investment experience than themselves. However, there is another interesting feature about this cluster. Although the overall proportion of early-stage deals is just 27% for the VC firms in this cluster – that for the FVCs is about 39% while that of their domestic counterparts is just about 8%. Similarly, the FVCs syndicate just 62% of their deals as compared to that of the DVCs of 92%. Thus, unlike what is conventionally known – the FVCs in this cluster do not necessarily shy away from the early-stage ventures, especially when compared to their local counterparts. Moreover, they are also not using syndication to get over the information asymmetry risks as extensively as is believed.

Thus, it becomes important to probe the other factors that enable these VCs to get over the underlying uncertainties. This could be mainly attributed to their foreign entry strategy. 100% of the FVCs in this segment have a physical office present in India – which can be viewed as an important mechanism of enhancing their level of social capital and accordingly facilitating access to local knowledge (Meuleman and Wright, 2011). Moreover, all these offices are led by executives of Indian origin – especially those with experience of working overseas. These professionals are likely to possess a superior insight into the cultural aspects of both geographies – the host nation and the investment destination. Consequently, such professionals play an important role in blending together 'the best of both worlds'. In fact, the VCs investing in India have since long followed the strategy of hiring their investment executives from other Indian VCs (Dossani and Kenney, 2002).

The VCs in this segment are also less likely to be early-stage specialists. This could possibly be explained as follows. Early-stage specialisation warrants a presence of professionals with a knack for identifying opportunities at very nascent phases (Patzelt et al., 2009). Especially in tech domains, the executives with prior founding experience in similar realms are better placed to understand the nuances of newly emerging businesses and technologies (Davidsson and Honig, 2003) especially the technological risks (Patzelt et al., 2009). Hence early-stage focused VCs often recruit erstwhile founders as partners on their investment teams. However, for this VC segment, the presence of investment executives with the above backgrounds is limited (13%) as compared to the overall average (27%). This factor possibly deters them from becoming early stage specialists.

7.1.3 Low syndication – high specialisation cluster

The VCs in this segment are early-stage specialists (about 85% of their deals have an early-stage focus as compared to the sample average of 47%) however, they demonstrate a very low intensity of syndication (just 14% of their deals are syndicated as compared to the sample average of 54%). Conventional understanding would find this highly counter-intuitive – since early investment stages involve high information asymmetry and therefore a need for greater resources which in turn needs to be overcome by syndication.

But one must understand that there exists a difference between the *need* to syndicate and the *opportunity* to syndicate (Dimov and Milanov, 2010). Syndication warrants the presence of contacts and networks for which the VCs require status and reputation (Lu et al., 2013). The latter comes from investing experience and the VCs in this segment seem to score low on this. These VCs are the youngest (3.5 years) and have also funded the least number of deals (11 deals) on an average. Moreover, they are also much smaller as compared to their other counterparts in terms of the size of their investment teams (team size of four as compared to the average of six). Also, this cluster also ranks the lowest in terms of the proportion of investment executives with prior experience of working for other VC firms in India and abroad (66.7% as compared to the overall average of 36%). Since, it is ultimately the network of contacts of the team members that facilitates syndication; the above factors seriously restrict their ability to do so. Among the FVCs in this segment, a very small proportion (only about 33%) has a physical office in India. This further reduces their connections with local networks and thus inhibits syndication.

The question that becomes important then is how these VCs deal with the information asymmetry, if it is not possible to syndicate. This can be answered in two ways. First, most VCs in this segment are of domestic origin (about 81%), wherein they possess deep understanding of specific local conditions and accordingly the underlying asymmetries (Devigne et al., 2013). Their networks with the local ecosystem make them adept in terms of understanding the underlying institutional mechanisms. Second reason that possibly enables them to compensate for the low syndication is their high sector focus – about 88% of the VCs are sector-focused and invest in only 3 sectors with a Herfindahl index of 0.694. The sector focus becomes an important resource resulting in strong ‘learning curve’ and ‘spillover’ effects that enable them to get over the uncertainties associated with VC investing.

7.1.4 High syndication – high specialisation cluster

The VCs in this segment demonstrate high early-stage specialisation and are also intense syndicators. The small size of financial resources (USD 30 Mn) and moreover their early-stage focus enhances their need for syndication. The composition of the human capital of this segment could be looked upon as an imperative facilitator of the syndication process. Their larger team size possibly enhances their access to networks. Moreover, this segment also has the highest proportion of VCs with erstwhile technology entrepreneurs on their investing teams (56%). This becomes a significant source of tacit knowledge – which facilitates both stage-specialisation and syndication (via their networks). The *niche* skill sets of their investment executives also help them to focus on high-tech domains. Accordingly, this segment has the highest proportion of VCs focused on technology domains (94%).

In terms of the social capital, although the VCs in this segment score low in terms of their experience (low age, only four years of operations in India), they seem to have found a novel way to compensate for the same. They achieve this by establishing offices at multiple locations in India. In fact, this segment has the highest number of VC firms with offices at multiple locations (25% as compared to the sample average of 19%). By doing so, they are likely to get an access to the local networks in each of these start-up clusters and keep abreast of the investment opportunities the very early-stages itself. This not only reinforces their early-stage specialisation focus, but by establishing connections with the VCs present therein, contributes towards enhancing their prospects of syndication as well. About 25% of the VCs in this segment have opened offices across multiple locations in India. Another factor that works in their favor is that a relatively large proportion of these VCs are in Bangalore – which again helps in facilitating their access to the underlying ecosystem and moreover the specialised human capital pool – both of which have an impact in terms of increasing the syndication and specialisation intensities.

7.2 Strategy interactions

Having analysed the profiles of the individual clusters, it is important to assess the interactions among the two strategies of syndication and specialisation. We define interactions in the following manner: two segments that seem apparently similar in terms of one strategy (e.g., high syndication) may significantly differ in terms of their relative usage of the other viz. early-stage specialisation. It is important to understand how and why such interactions exist and how the underlying resource composition impacts the same.

The VCs in clusters 3 (low syndication – high specialisation) and 4 (high syndication – high specialisation) are both highly specialised in terms of their early-stage investment focus, and yet the former exhibits a low intensity of syndication while the latter a higher intensity of the same. The main difference among the two seems to be in terms of their endowment of social capital (as seen from the composition of their investment teams, physical presence in a vibrant start-up cluster (viz. Bangalore) or the presence of their offices at multiple locations). In general, VCs with an ample endowment of social capital seem to have access to more opportunities that facilitate the syndication process.

Similarly, although clusters 1 (low syndication – low specialisation) and 2 (high syndication – low specialisation), both display low specialisation intensity, the VCs in the

former syndicate with low intensity and vice-versa. For cluster 1, the abundant endowment of their financial and human resources coupled with their investment focus on more conventional domains, seems to lower the need for syndication. Here, the social capital in terms of the investment team composition or the physical location of the VC firm does not seem to matter much as they already have high status and reputation as indicated by their age of operations in India. For cluster 2 however, syndication is important mainly owing to the large FVC presence therein. They consciously overcome these uncertainties by strategically enhancing the level of their social capital – mainly by their foreign entry strategy (establishment of physical offices) and choosing to co-locate with other vital stake-holders in Bangalore.

Similarly, clusters 2 (high syndication – low specialisation) and 4 (high syndication – high specialisation) both syndicate intensely, but differ sharply in terms of their early-stage specialisation focus. For cluster 2 only about 27% of the deals have an early-stage focus, whereas for cluster 4 it is as high as 87%. The main difference among the two clusters seems to be in terms of the composition of human capital. Although, the size of the respective investment teams is comparable, the VCs from cluster 4 have a strong presence of investment teams with prior founders. The *niche* skill sets of their investment executives result in further reinforcing of stage and sector specialisation. Among the VCs in clusters 1 (low syndication – low specialisation) and 3 (low syndication – high specialisation) – both of which syndicate less intensely, the one that is highly specialised exhibits a human resource composition that is sharply different from the other. Although the investment teams of the VCs in cluster 3 are much smaller in size, an overwhelmingly large proportion of these have the presence of erstwhile founders.

To conclude, VCs that are apparently similar in terms of their syndication intensities, differ with respect to the usage of specialisation strategies – mainly owing to the diverse composition of their human capital. Similarly, the VCs that appear similar in terms of their specialisation propensities differ with respect to syndication intensities mainly because they are differential endowment of social capital. Interpretation with respect financial capital is more straightforward wherein – lower fund-sizes are associated with greater syndication but a higher early-stage specialisation.

8 Conclusions, managerial and policy implications

In this paper, we use the RBV to analyse the intensity of syndication and stage specialisation – as risk management strategies for the VC firms investing in India. To start with, we hypothesise the intensity of strategy usage to be driven by three main resource-related attributes: pre-existing resource structure of the VC firms, the need to access greater resources and the ease of access to resources. Further, by using the K-means clustering algorithm, we divide the VCs into four clusters (low syndication and low specialisation, high syndication and low specialisation, low syndication and high specialisation and high syndication and high specialisation), and then review how these attributes influence the variations in strategy usage and the underlying interactions therein.

By analysing the three main resource types (viz. the financial, human and social capital) we draw important conclusions pertaining to the intensities of syndication and specialisation. In general, the social capital seems to impact the degree of syndication

while human capital the extent of stage specialisation. Large financial capital seems to reduce the need for syndication, and is also found related to a lower level of specialisation. Moreover, the location characteristics of the VC firm (such physical location in a start-up cluster, offices at multiple geographic locations and presence of a physical office at the investment destinations) are likely to influence the social capital and accordingly the intensity of strategy usage. Likewise, sector-focus (particularly high-tech focus) impacts specialisation but not necessarily syndication. We further review the interactions among these strategies, wherein we show how the firms that overtly appear similar in terms of their usage of one strategy could significantly differ in terms of the intensity of usage of the other. We relate these interactions to the underlying resource composition especially the human and social capital.

We trust, that this study makes many important contributions to the extant literature in the across multiple domains. First, we contribute the literature on VC investment strategies. Although, syndication and specialisation has been separately reviewed so far (Hochberg et al., 2007; Gompers et al., 2008; Dimov and De Clercq, 2006; Hochberg et al., 2015a, 2015b), the mutual interactions driving the joint usage of the two has not been adequately analysed. By probing how the underlying interactions between the levels and relative composition of human and social capital drive apparently similar VCs to differ in terms of their relative intensities of syndication or specialisation, we add a new dimension to the existing studies. This study brings out the principal factors in emerging economies like India hinder specialisation – especially sector specialisation. The most important among them is the low depth of the market for viable deals. Although, several efforts have been made to address this issue, only a few have been successful. The government policies have encouraged the establishment of academic incubators/business accelerators towards this end, yet only a few of them were found to be viable (Gerlach and Brem, 2015; Aaboén and Löfsten, 2015). This is because, most of these efforts were focused on facilitating access to mere ‘physical resources’ and not ‘networks’, when networks are the key to VC investing.

Secondly, lack of sufficient expertise and resources in scaling up companies possibly drives most DVCs to focus on early-stage specialisation rather than sector specialisation. This can be deciphered from the fact that the most prominent exit routes for most DVCs are often mergers and acquisitions or secondary sale to an upstream VC firm (VC Circle, 2014).

The study also brings out novel strategies undertaken by FVCs to enhance their social capital in emerging economies. Factors such as the physical location of their firm, establishing offices across multiple locations and their foreign entry strategy has impact on their social capital and accordingly the intensity of strategy usage demonstrated by them.

Next, we contribute to the literature on human capital of VC firms. While, the composition of human capital has been extensively analysed especially in the context of other entrepreneurial firms (Krishnan et al., 2006; Lee and Park, 2008), the same cannot be said of VCs. In fact, VCs may be looked upon as a special kind of entrepreneurial firm wherein the opportunity recognition skills of partners play a critical role in identifying, funding and growing the nascent ventures. This study reveals how the specific backgrounds of the VC firm professionals provide a proxy for their niche skill sets that guide the nature of strategies pursued by them.

Lastly, we extend the literature on emerging economies to India. Among the emerging economies, only the syndication strategies of the VCs in China have been

briefly studied so far (Krishnan et al., 2006; Lee and Park, 2008). However, even these studies do not specifically link the same to the underlying resource structure of the VC firms. By linking syndication intensity to the resource composition and by explicitly analysing the specialisation and also the underlying strategy interactions, this study we believe takes the existing literature forward.

Our study has important managerial implications and policy conclusions. For VCs, it provides insights into understanding their resource structure and how they can compensate for their lack of the same. It provides practical guidelines about enhancing the levels of social capital – e.g., locating their offices in geographies with a vibrant start-up ecosystem is one of the ways to overcome the impediments associated with a lack of social capital. Establishing offices at multiple locations is another mechanism of establishing connections with the local networks. For FVCs establishment of local offices is an important channel of overcoming their initial low endowment of social capital. The study has implications for enhancing the composition of human capital as well. For VCs intending to be domain-specialised, the study highlights the importance of hiring their investment professionals from specific backgrounds viz. erstwhile founders; while for VCs that are ‘generalists’ it stresses on the need of hiring executives with prior VC investing experience.

For policy makers, as well, our study provides important guidelines. Syndication and specialisation are both important for risk management and the VCs should be provided with an environment wherein they can practice both these strategies effectively. Syndication can be facilitated by establishing a vibrant ecosystem and government has a key role to play in this regard by providing a supportive policy environment such that it encourages the establishment of a greater number of such start-up clusters. Specialisation is important to spot nascent opportunities, however that requires the existence of a viable deal flow. Incubators and accelerators can be regarded as the vital sources of such early-stage deals. In this regard, the policies need to focus on not just establishing more such set-ups but also in terms of making the existing ones truly functional.

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